Relational Geometry Program: from difference to own closure

Ontological foundations, mathematical realizations, relational time and open lines of research

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October 24, 2026

English translation of `Programa_Geometria_Relacional_Manuscrito_Publico_2026-10-24.md`. The Spanish text is the authoritative version. A glossary of the program's terms is in `01_HILO/HILO_ONTOLOGICO_INTEGRADO_2026-10-24.en.md`.

Abstract

The **Relational Geometry Program** asks whether a minimal economy of differences, reflections, references and own closures can organize a common language for studying mathematical, computational and physical phenomena. Its starting point is neither a spatial metric nor a universal time: it is a unit with no certified internal differences, the possibility of reflection, and the need to keep a difference when a structure is recognized from another reference. The proposal is developed in separable layers: ontological narrative; formal definitions; results proved under explicit hypotheses; reproducible trials; and physical bridges still subject to testing.

This edition brings together a result that was scattered: **measured time is understood as the comparison of rhythms of recurrence with respect to a cycle chosen as reference**. The moving mirror represents the change of focus, orientation or scale from which that comparison is made. The own closure of each singularity requires neither equal rhythms nor the disappearance of its differences. The models MF-43 and O45 give exact mathematical realizations of these principles under declared conditions; identifying them universally with physical time and gravity remains a hypothesis open to investigation.

Scope and position: this text presents an open program with results that can be verified within specified models. It does not claim to solve P versus NP, the global regularity of Navier–Stokes, the Yang–Mills mass gap, or an experimentally confirmed theory of gravity.

1. The thread: unit, reflection, difference and closure

The ontology proposes starting from a totality that cannot yet be compared with anything else. Reflection introduces a distinction and its reference. As it repeats, some relations produce references sufficient to be recognized as units; others keep indeterminacies that require the language to be extended.

The central vocabulary keeps these boundaries:

- **Novelty**: a question or difference that the previous signature cannot decide without refinement.
- **Normalization**: keeping a relation when the reference changes; it does not imply that two physical dynamics are equivalent.
- **Own closure**: an identity that can be sustained with respect to a contract of comparison. It is not the same as knowing every inner state.
- **Commons**: shared relations and interfaces that allow structures to be compared without declaring them identical.

Language guides the formalization, but the metaphors of the mirror, the cube and the spheres do not replace mathematical proofs or physical measurements.

2. Exact realizations and laboratories

The arithmetic realization of R-01 makes it possible to express the same relation under normalizing multiples: for a multiple L of N, $h/N = (hL/N)/L$ is preserved. If a signature modulo N is extended by a prime p that does not divide N, the common reference $L = Np$ recovers both signatures. This is a conditioned arithmetic result; by itself it does not define a physical flow.

The **mirror cube laboratory** studies reflections, orbits and closures. CME-003 represents four references by parity comparisons that form a regular tetrahedron under a chosen Euclidean metric. The elementary exchanges of pairs of references produce pairs of mirrors with chamber angles $\pi/2$ and $\pi/3$; when the family of generators is extended by sign changes, pairs at $\pi/4$ appear. The orders of composite elements are carefully distinguished from the orders of the product of two elementary mirrors. The choice of that metric realization, and the ontological step toward the four references, require further justification.

The **transparent spheres laboratory** studies points of tangency and possible contact routes. Contact topology, ideal tangential transport and the real optics of glass are different contracts. The circuit of four spheres, or C4, is investigated as an example of a relation that allows an alternating mode; it is not automatically identified with a particle or with a real fluid.

3. Time: the rhythm at which processes happen

[O] Ontological reading. The base cycle represents the *reference*. Each singularity is sustained by its own closure and shows recurrences relative to that reference. Measured time does not precede these relations: it expresses the comparison of the

rhythms at which processes happen. A coincidence of phase does not erase the particular identity, nor does it force interactions to disappear.

[T conditioned] MF-43: recurrence without an external clock. For an amplitude on a homogeneous cycle, with a local, symmetric and reversible update, one studies

$$x_{j,n+1} - 2x_{j,n} + x_{j,n-1} = -\eta(L_a x_n)_j.$$

An eigenmode with eigenvalue λ_m admits, under the stability condition $0 \leq \eta\lambda_m \leq 4$, a phase per update

$$\boxed{\Omega_m = 2\arcsin\left(\frac{\sqrt{\eta\lambda_m}}{2}\right)}.$$

Its logical period is $P_m = 2\pi/\Omega_m$, if $\Omega_m > 0$. Two modes of the same contract satisfy $P_i/P_j = \Omega_j/\Omega_i$. Under the additional condition of non-dispersive propagation on the branch studied, MF-43 selects $\eta = 1$. Neither n nor Ω_m is yet equivalent to seconds or hertz.

[T conditioned] O45: comparison from the moving mirror. If Θ_0 is an accumulated reference phase, defined physically or formally, and Θ_i is that of a singularity, then, wherever the denominator does not vanish,

$$\boxed{\Gamma_{i|0} = \frac{d\Theta_i}{d\Theta_0} = \frac{\kappa_i}{\kappa_0}}.$$

The rates κ must be computed in a fixed reference, or through a connection that removes arbitrary changes of the phase zero. A relative clock counts the cycles of Θ_0; this is invariant under an admissible reparametrization of the auxiliary index. Three rhythms, of flow, singularity and particle, produce two independent differences, provided their phases are defined and comparable.

[T conditioned] Radius and rhythm. A second comparator, continuous and spectral, defines $R_i = \lambda_i^{-1/2}$ and $\omega_i = \sqrt{\Gamma\lambda_i}$. In that model one derives $R_i\omega_i = \sqrt{\Gamma} =: v_*$. If the radii keep proportions $n_i = R_i/R_0$, then $\omega_i/\omega_0 = 1/n_i$. This ω must not be replaced, without proof, by the discrete phase Ω_m of MF-43. Identifying $v_* = c$ needs a physical hypothesis about universal causal propagation, and its dimensional calibration.

The statement "c is the base rhythm" is therefore understood here with care: c is **the candidate limiting velocity associated with spatial advance per reference cycle**, not a frequency with units of inverse seconds.

4. Identity, distance and gravity: three different levels

Identity is the own closure that a structure keeps. **Distance** compares separations or scales of two identities through a reference. **Gravity**, if the program manages to derive it, must be a dynamical law that modifies physical relations while keeping the admitted closures. Defining a distance is not the same as producing a force.

GR3 proves, on a binary K4 lattice, that global closure can exist without all labels being equal. Unison concordance is a reconstructible compatibility of differences; it does not require *strong unison*, with zero differences. This matters: two systems can share a reference rhythm and keep different closures.

A candidate energy $V(r, \delta) = K(r)(1 - \cos\delta)$ models a penalty for phase mismatch. Since $V(r, 0) = 0$ for every separation r, its radial gradient also vanishes when $\delta = 0$, if it is the only term. **It cannot be used by itself to explain a persistent gravitational interaction between perfectly synchronized systems.** This does not refute every phase-dependent gravitational model; it delimits the reach of this expression.

The program's path is to investigate a dynamics of relations between closures, scales and relative rhythms, not to replace $1 - \cos\delta$ by a $1/r$ potential without proof, nor to define an acceleration from a metaphor. The corpus keeps hypotheses of gravitational tuning and conditioned field realizations; still pending are a non-circular selection of the physical coupling, the quantitative recovery of observables, and an independent test against relativity and experiments.

5. Three branches of contrast

Navier–Stokes. Branch O93 contains balances, normalizations and continuation criteria that are exact under its contract. Certificates for subclasses were obtained, along with falsifiers that prevent closing the dynamics with insufficient information. General three-dimensional global regularity without external force is not proved by these results.

P versus NP. LAB-CONV v2 investigates boundary signatures and elimination costs, with computational verification. The barrier N1 itself shows the limits of the table engine under defined contracts. It proves neither $P = NP$ nor $P \neq NP$.

Yang–Mills. R-02 studies pairs of SU(2) loops and their invariants; the loss of the commutator under certain blockings shows that a signature can discard relevant differences. It does not establish the existence of four-dimensional quantum Yang–Mills with a uniform gap.

These three branches are places where the thread is tested, not axioms that ground the ontology after the fact.

6. What a reader can verify

The package accompanies this manuscript with its integrated thread, the realizations CME-001 to CME-003, the sphere and mirror laboratories, ROSI programs and orig-

inal sources. To check integrity, run `python3 verificar_integridad.py` from the root folder after unpacking the edition. The reproducible realizations come with their own verifiers; numerical results must be interpreted within their samples and assumptions.

Epistemological levels: [O] ontological interpretation; [D] rule or definition; [T] theorem under explicit hypotheses; [E] reproducible trial; [F] falsifier or counterexample; [H] physical hypothesis; [A] open obligation. A result already known in the literature can be found again by the program without thereby becoming an original theorem of physics.

7. Conclusion and research horizon

The consolidated proposal holds that novelty requires relations to be refined, that some closures admit a common representation without eliminating all their differences, and that the recurrences of such closures make it possible to compare rhythms. **The base cycle keeps the reference; the multiples distinguish recurrences; time quantifies that comparison.** The cube and sphere laboratories provide images and models that make certain operations precise. The relation to gravity, mass and physical observables remains a matter of derivation and falsification, not an established physical equivalence.

The invitation is to rebuild the arguments, run the tests, find the limits and propose new relations, without replacing the ontological thread by external analogies.

Sources and traceability within the package

As listed in the original, followed by their location in this package (translator's note).

1. **Integrated ontological thread**, chapter VII: reading and matrix of obligations. — `01_HILO/HILO_ONTOLOGICO_INTEGRADO_2026-10-24.md`, Part VII; in English, `.en.md`.
2. **Supplement on the base cycle, rhythms and gravity** (folder `01_HILO/SUPLEMENTOS`): derivations and contracts. — `01_HILO/SUPLEMENTOS/Suplemento_Ritmo_Ciclo_Base_Tiempo_y_Gravedad_2026-10-24.md`; in English, `.en.md`.
3. **MF-43, From mode to rhythm** (recovered sources): logical period and non-dispersive condition. — *Not included in this package.* The formula was checked again for this edition.
4. **Rhythm, radius and relational relativity** (recovered sources): the spectral product $R\omega$. — *Not included in this package.* The result also appears in chapter 9 of `01_HILO/imagen_total_3_9/Imagen_Total_3_9_Informe_General.md`.
5. **GR3, Unison concordance** (recovered sources): focus, closure and persistent differences. — `01_HILO/fuentes/07_ritmo_y_gravedad_recuperados/SUPLEMENTO_GR3_AJUSTE_UNISONO.es.md`.
6. **O45, Time and the moving mirror** (historical archive): covariant comparison of phases. — `05_ARCHIVO/navier_stokes_039_053/documentos/045_TIEMPO_ESPEJO_MOVIL_RITMOS_NS_2026-10-20.md`.
7. **MF-44B, Causal normalization** (continuity sources): relational limit per update. — `01_HILO/fuentes/06_hilo_del_dos/continuidad_2026_10_14/FUENTES_COTEJO/MF44B_normalizacion_causal_cierre_global_v1.md`.

8. **CME-003**, realization of four references and mirrors; see the editorial erratum in `00_PUBLICACION`. — `02_REALIZACIONES/01_cubo_de_espejos/CME-003_de_las_cuatro_referencias_a_los_angulos/`. The erratum is not included in this package; the distinction between the orders of composite elements and those of products of two mirrors is now stated and checked in CME-003 itself.

Further notes for this package (translator):

- Section 6 mentions ROSI programs. This package contains ROSI's description (chapter 10 of edition 3.9 and `01_HILO/fuentes/04_sintesis_integral_2026-10-20/07_ROSI_LENGUAJE_EXPERIMENTOS.es.md`), not its programs.
- The no-go result of section 4 was already written on October 8, 2026: `01_HILO/fuentes/07_ritmo_y_gravedad_recuperados/04_SINCRONIZACION_ESCALAS_CONTRATO_Y_NO_GO.md`.

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